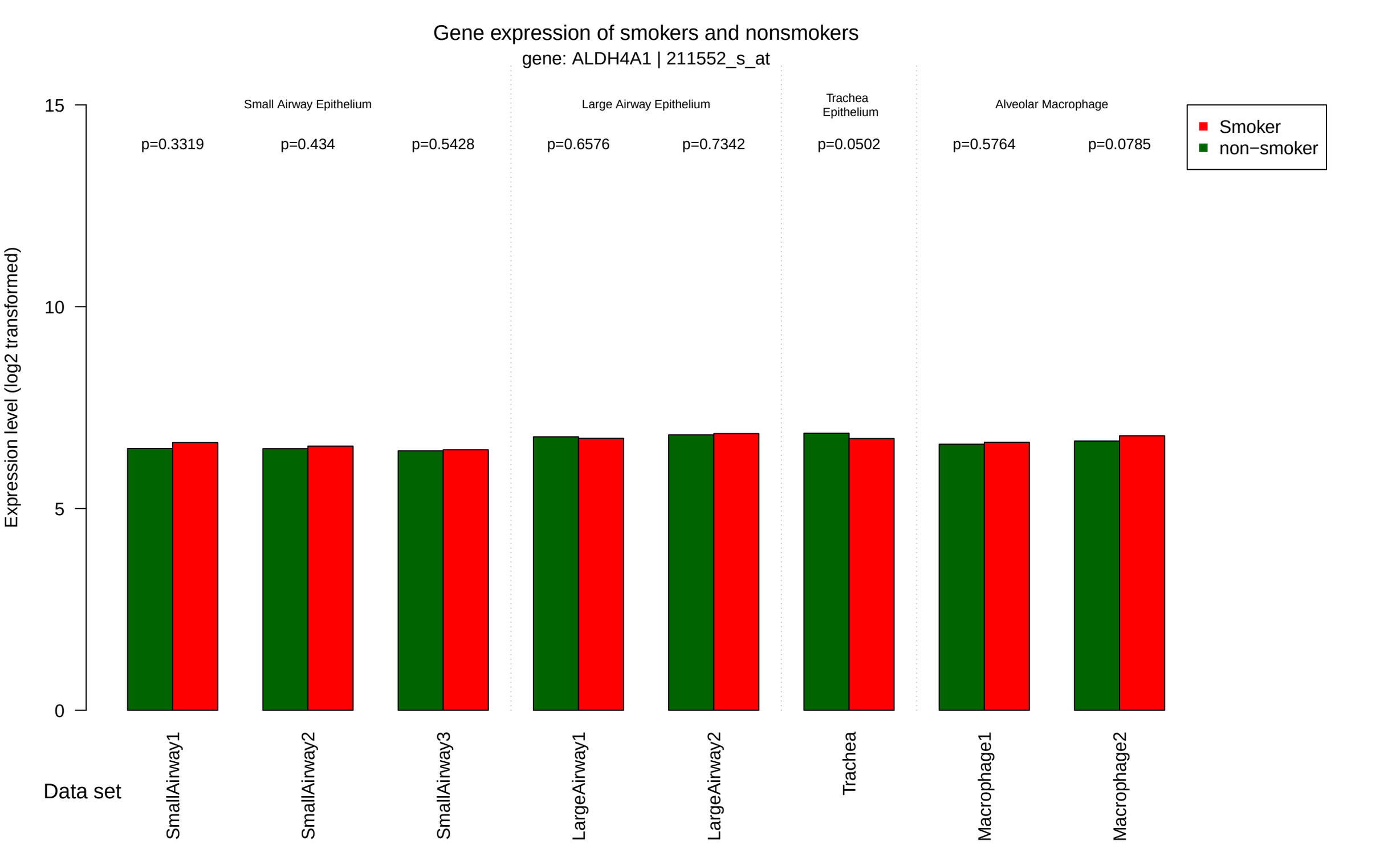
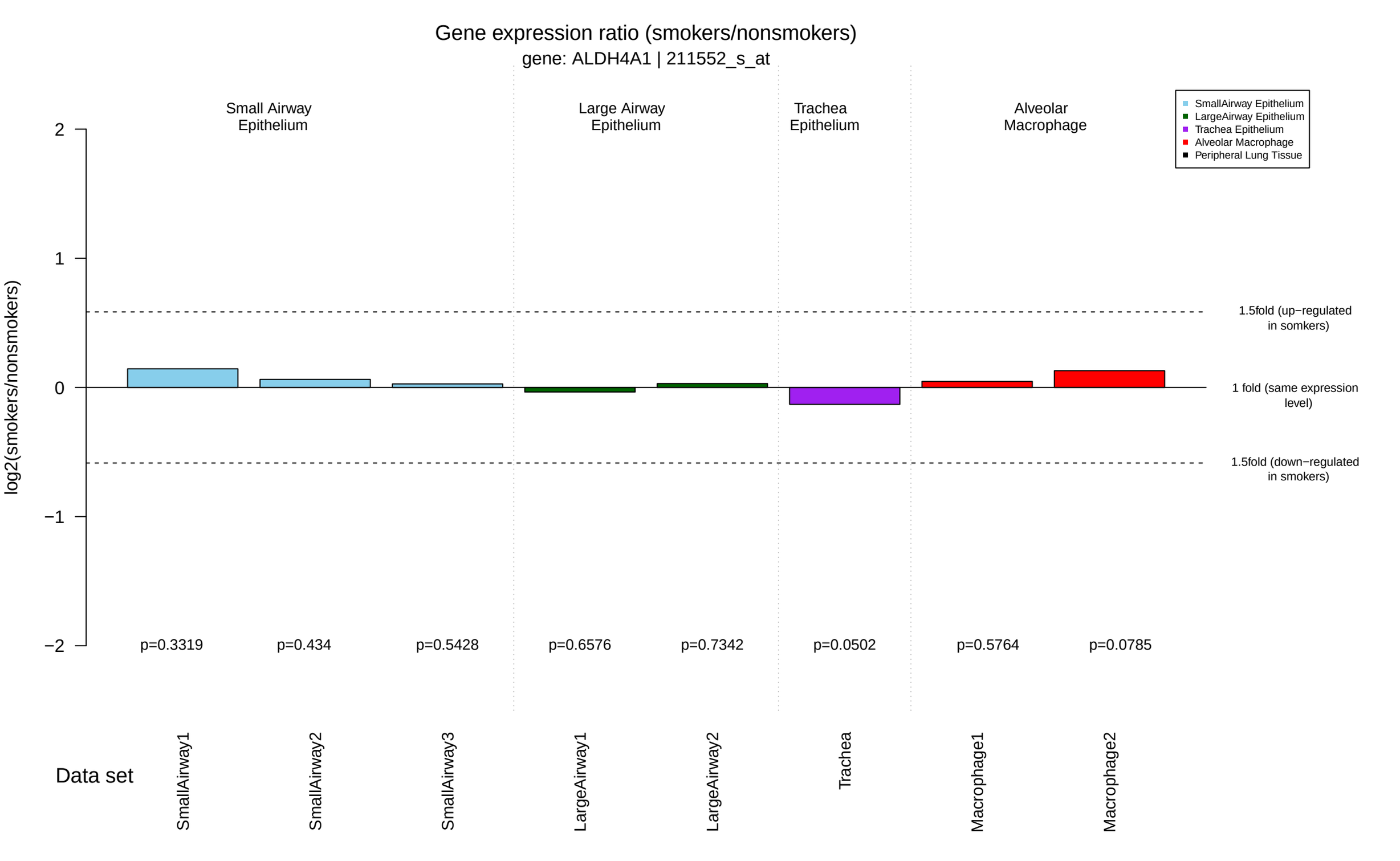
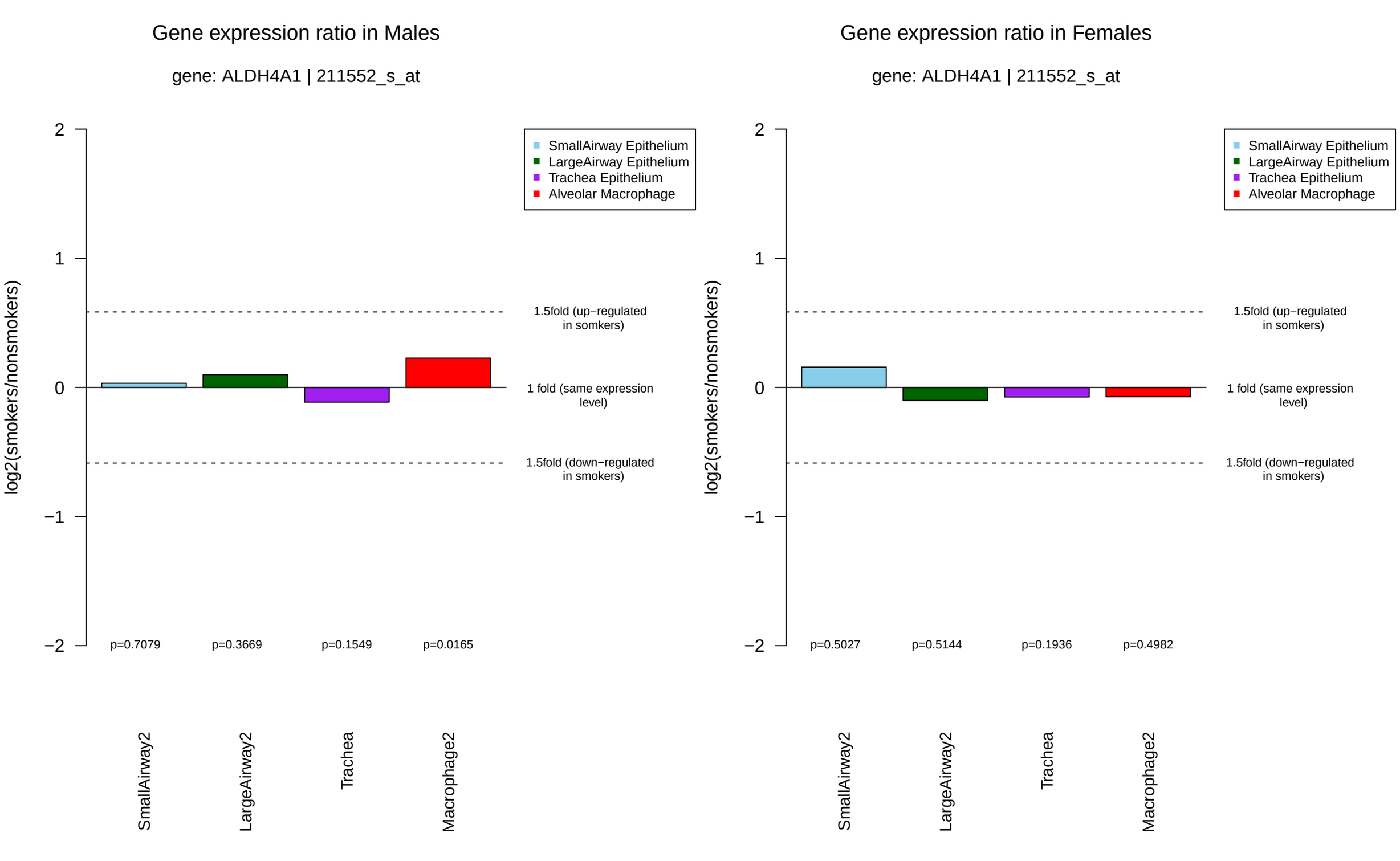
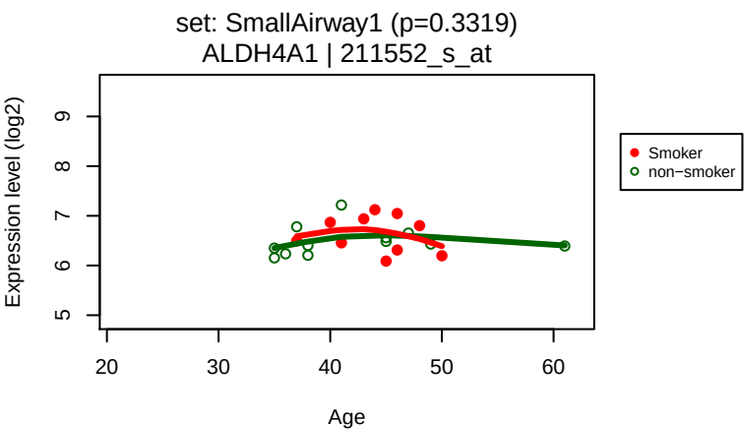
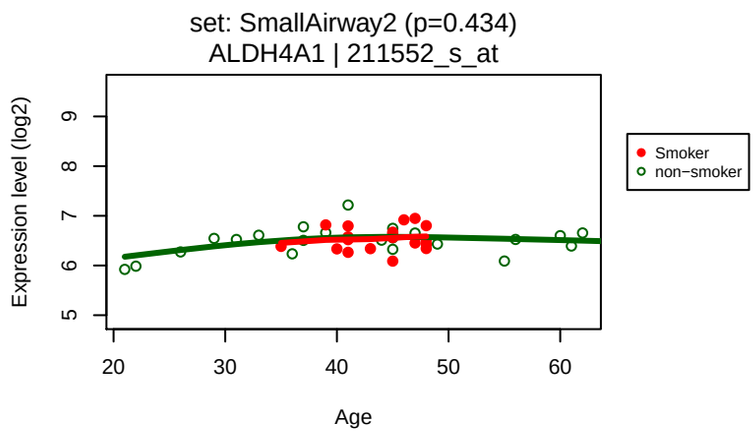
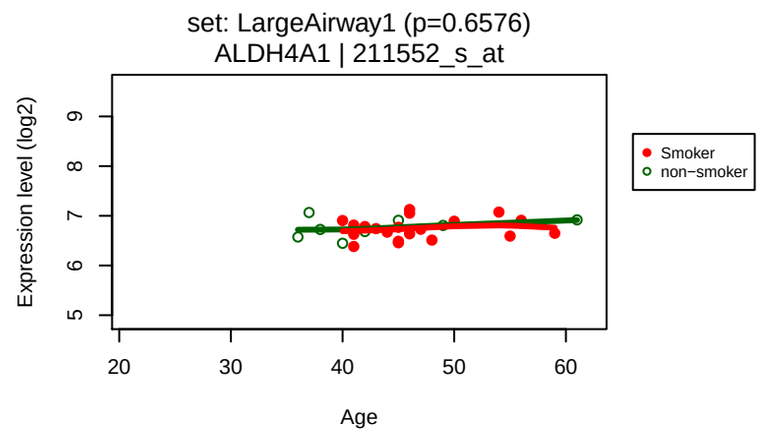

Gene expression of smokers and nonsmokers
gene: ALDH4A1 | 211552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3319
p=0.434
p=0.5428
p=0.6576
p=0.7342
p=0.0502
p=0.5764
p=0.0785
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

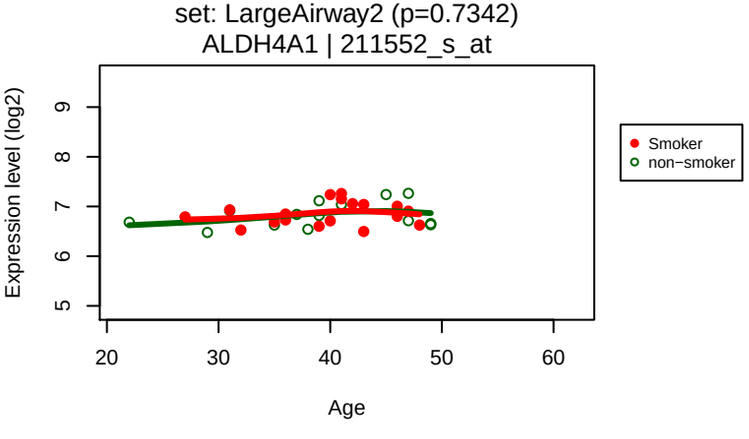

Gene expression ratio (smokers/nonsmokers)
gene: ALDH4A1 | 211552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3319
p=0.434
p=0.5428
p=0.6576
p=0.7342
p=0.0502
p=0.5764
p=0.0785
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

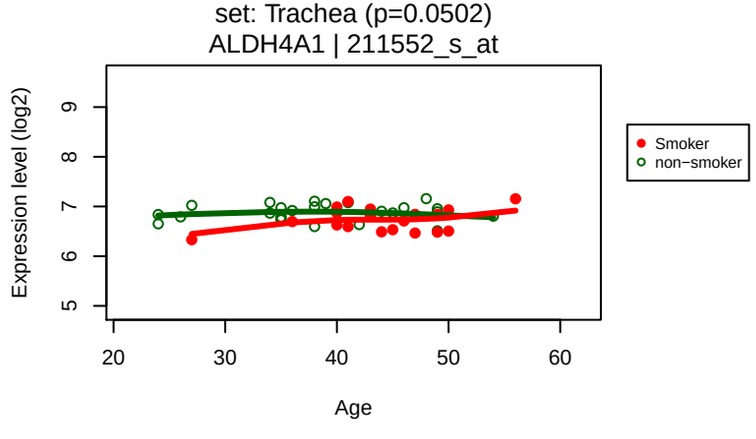

Gene expression ratio in Males
gene: ALDH4A1 | 211552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7079
p=0.3669
p=0.1549
p=0.0165
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH4A1 | 211552_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5027
p=0.5144
p=0.1936
p=0.4982
SmallAirway2
LargeAirway2
Trachea
Macrophage2

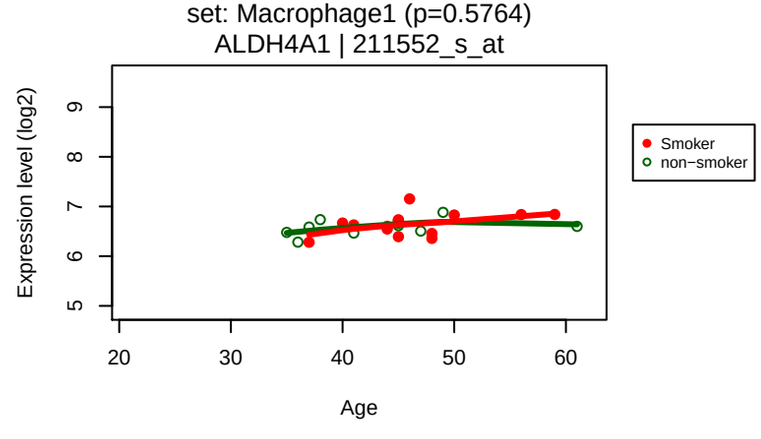

set: SmallAirway1 (p=0.3319)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker

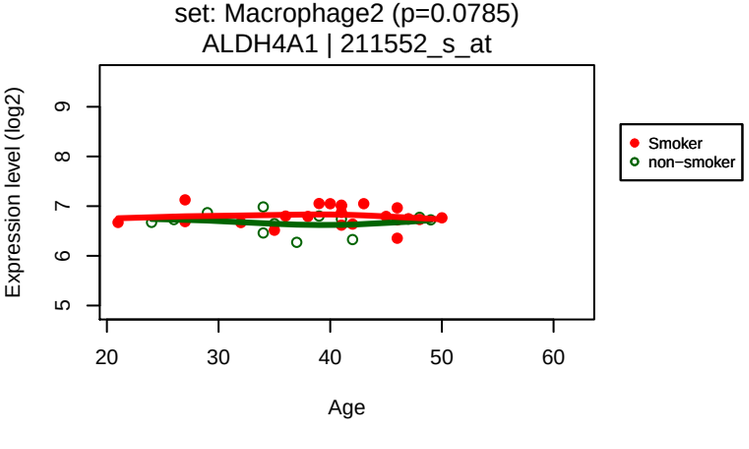

set: SmallAirway2 (p=0.434)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker

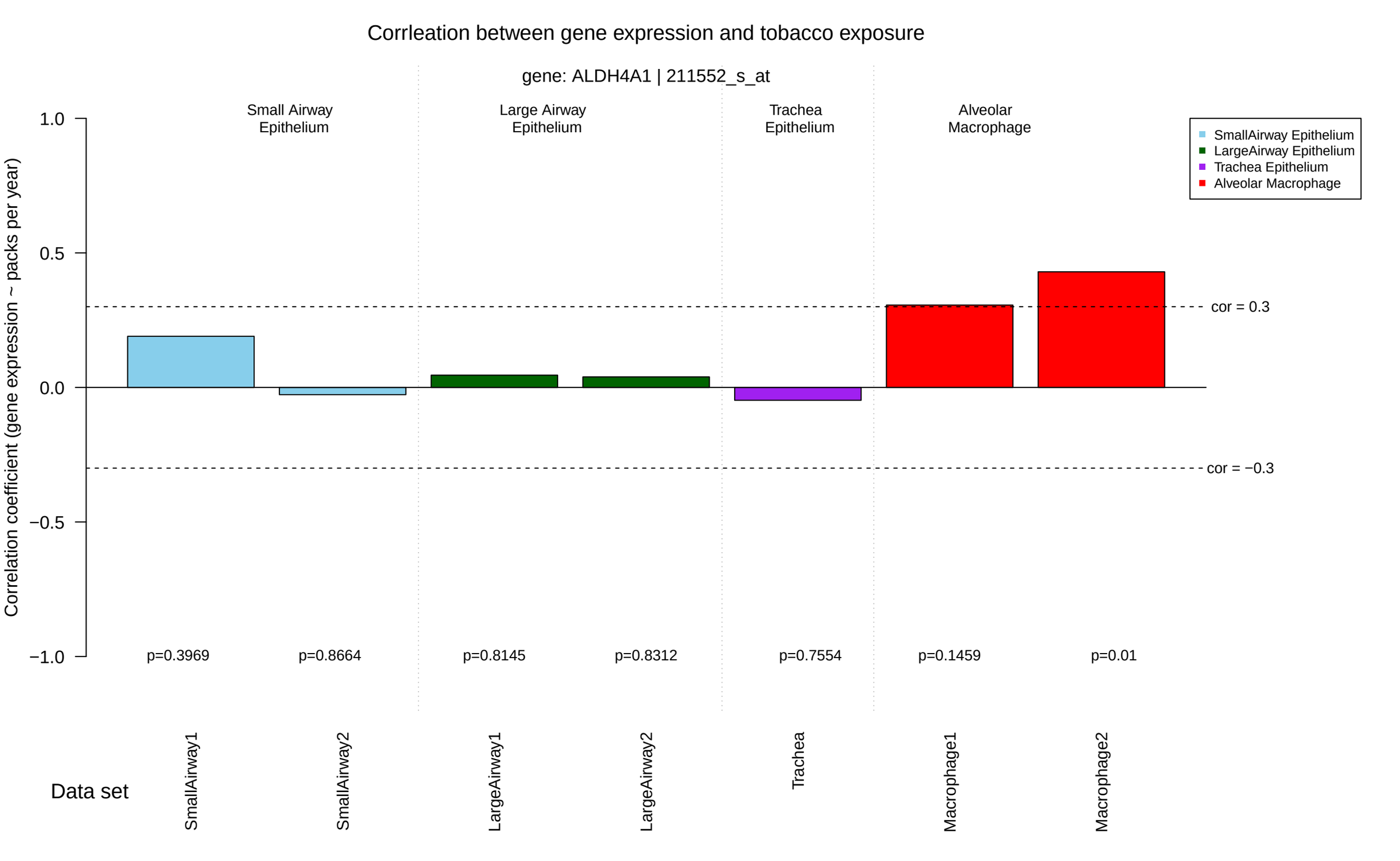

set: LargeAirway1 (p=0.6576)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7342)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0502)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5764)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0785)
ALDH4A1 | 211552_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH4A1 | 211552_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3969
p=0.8664
p=0.8145
p=0.8312
p=0.7554
p=0.1459
p=0.01
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2